12. Data Science Methodologies

"Things get done only if the data we gather can inform and inspire those in a position to make a difference"- Michael Schmoker

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

Motivation



Survivorship Bias

Motivation

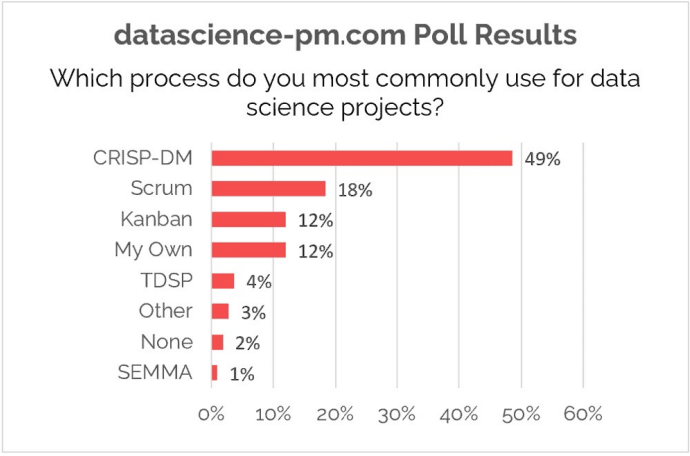


Surv

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Motivation

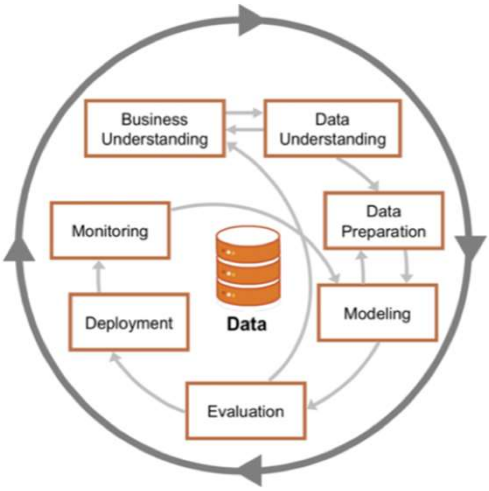


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CRISP-DM (CRoss Industry Standard Process for Data Mining)

- CRISP-DM is one of the most widely used methodologies for data science and analytics projects.
- Provides a structured approach across six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.



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CRISP-DM (CRoss Industry Standard Process for Data Mining)

- **Advantages:** It's versatile, industry-agnostic, and promotes iterative development, allowing for adjustments as more insights are gathered.

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TDSP (Team Data Science Process)

- The TDSP is an agile and iterative data science methodology that you can use to deliver predictive analytics solutions and AI applications efficiently.
- The TDSP lifecycle consists of five major stages that your team performs iteratively. These stages include:
  - Business understanding
  - Data acquisition and understanding
  - Modeling
  - Deployment
  - Customer acceptance

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TDSP (Team Data Science Process)

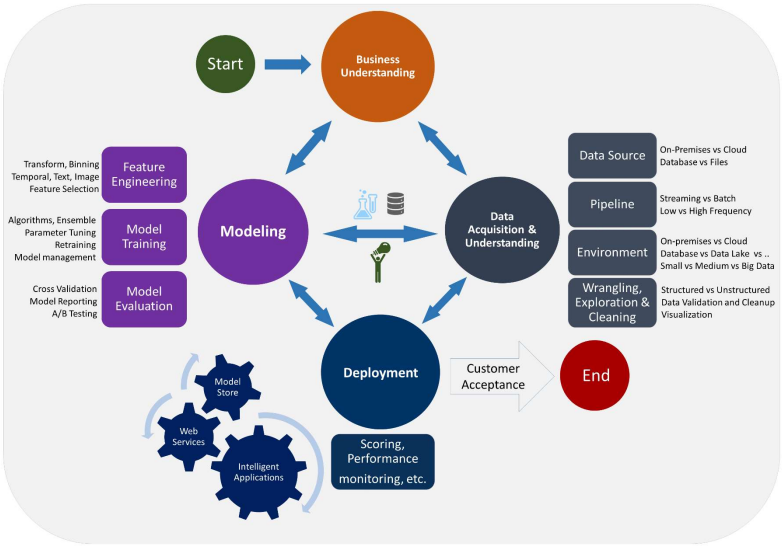
- These tasks and artifacts align with project roles, such as:
  - Solution architect
  - Project manager
  - Data engineer
  - Data scientist
  - Application developer
  - Project lead

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TDSP (Team Data Science Process)

Data Science Lifecycle



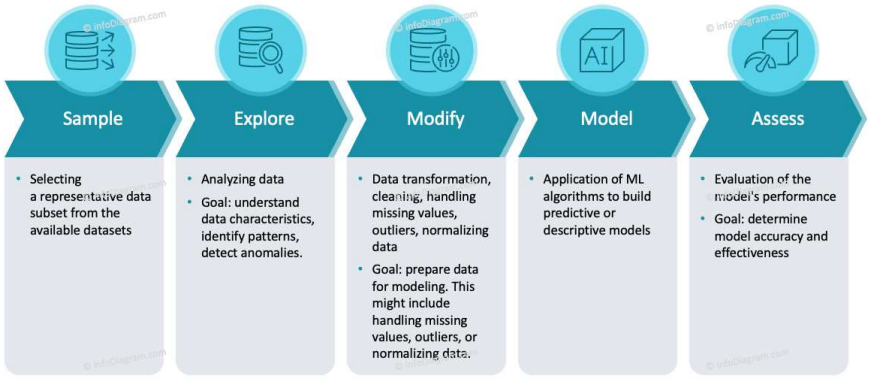
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SEMMA (Sample, Explore, Modify, Model, Assess)

SEMMA Methodology of Data Mining Process

Sample Data, Explore, Modify Data, Model building, Assess



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